

Chris Agia

cagia@cs.stanford.edu
www.chrisagia.com
github.com/agiachris
linkedin.com/agiachris

EDUCATION

Stanford University 2021 – 2026
Ph.D. in Computer Science, AI & Robotics
Co-advisors: Prof. Jeannette Bohg, Prof. Marco Pavone
Research: planning & decision making, deployment reliability & interpretability, data curation [Thesis] [Talk]
University of Toronto 2016 – 2021
B.A.Sc. in Engineering Science, Robotics
Advisor: Prof. Florian Shkurti. Graduation with Honors, Dean's List 2018 – 2021
Research: learning for robot perception, planning, and decision making [Thesis]

SELECTED RESEARCH EXPERIENCES

Autonomous Systems Lab, Stanford University Stanford, CA, USA
Graduate Researcher advised by Prof. Marco Pavone 2022-03 – 2026-03
Topics: (focus) reliability for imitation learning [Site, Site], *foundation models for robot safety* [Site, Site, Site, Site]; *(involved) uncertainty quantification, out-of-distribution detection, deep offline reinforcement learning*
Interactive Perception and Robot Learning Lab, Stanford University Stanford, CA, USA
Graduate Researcher advised by Prof. Jeannette Bohg 2022-01 – 2026-03
Topics: (focus) robot data curation [Site], *task and motion planning with learned skills* [Site] *and large language models* [Site, Site, Site]; *(involved) imitation learning, deep reinforcement learning, constrained optimization*

RESEARCH EXPERIENCES

FAIR Embodiment & Actions, Meta Menlo Park, CA, USA
Research Scientist Intern advised by Dr. Krishna Jatavallabhula and Dr. Franziska Meier 2025-06 – 2025-12
Topics: large-scale world modeling for robotics
NASA Jet Propulsion Laboratory, California Institute of Technology Pasadena, CA, USA
Visiting Researcher advised by Dr. Issa Nesnas and Dr. Saptarshi Bandyopadhyay 2023-06 – 2023-09
Topics: autonomy for deep space exploration [Site]; *Section 347: mobility and robotic systems*
Stanford Vision and Learning Lab, Stanford University Stanford, CA, USA
Graduate Researcher advised by Prof. Jiajun Wu 2021-09 – 2022-02
Topics: neuro-symbolic models for AI task planning
Robot Vision and Learning Lab, Vector Institute & University of Toronto Toronto, Canada
Undergraduate Researcher advised by Prof. Florian Shkurti 2020-05 – 2021-05
Topics: robot task planning in structured world models with graph neural networks [Site]
Mobile Robotics Lab, MILA & McGill University Montreal, QC, Canada
Research Intern co-supervised by Prof. Gregory Dudek and Prof. David Meger 2020-01 – 2020-05
Topics: learning for visual SLAM [Paper], *visual representations for reinforcement learning* [Paper]
Noah's Ark Lab, Huawei Research Canada Markham, ON, Canada
Deep Learning Research Intern, perception and localization with Dr. Bingbing Liu 2019-05 – 2020-05
Topics: 3D semantic scene reconstruction [Paper], *segmentation and semantic SLAM* [Paper]
Autonomous Systems and Biomech. Lab, University of Toronto Toronto, ON, Canada
Research Intern supervised by Prof. Goldie Nejat 2018-05 – 2018-08
Topics: sim2real transfer for robot reinforcement learning [Paper]

INDUSTRY EXPERIENCES

Mixed Reality and Robotics, Microsoft	Redmond, WA, USA
Software Engineering Intern on the Scene Understanding and Data Teams (HoloLens)	2021-05 – 2021-08
<i>Topics: multi-agent reinforcement learning in mixed reality environments</i>	
Cloud, Google	San Francisco, CA, USA
Software Engineering Intern building ABI simulators with the Istio Networking Team	2020-05 – 2020-08

HONORS AND AWARDS

Best Conference Paper Award , Robotics: Science and Systems (RSS)	2024
Best Conference Paper Award , Intl. Conference on Robotics & Automation (ICRA)	2024
Deep Tech Fellowship, Ph.D. , Clear Ventures	2023
School of Engineering Fellowship , Stanford University	2021
First Prize, Programming , Toronto District & Ontario Provincial Engineering Competition	2019
Undergraduate Research Award , Natural Sciences and Engineering Research Council of Canada	2018
President's Scholars of Excellence Program , University of Toronto (<1%)	2016

PUBLICATIONS

Ordered by recency. *Representative works* are highlighted. [†]Equal contribution. Details in [Google Scholar](#).

Refereed Publications

- [1] **Diversity You Can Actually Measure: A Fast, Model-Free Diversity Metric for Robotics Datasets**
Conference on Robot Learning (CoRL), 2026 (**Oral Presentation, <3.1%**)
S. Sirigiri, N. de Lara, **C. Agia**, F. Shkurti, and F. Ramos
<https://arxiv.org/abs/2603.11634>
- [2] **Preventing Robotic Jailbreaking via Multimodal Domain Adaptation**
International Conference on Robotics and Automation (ICRA), 2026
F. Marchiori, R. Sinha[†], **C. Agia**[†], A. Robey[†], G. J. Pappas, M. Conti, and M. Pavone
<https://arxiv.org/abs/2509.23281>
- [3] **CUPID: Curating Data your Robot Loves with Influence Functions**
Conference on Robot Learning (CoRL), 2025
RSS 2025 Workshop on Robot Evaluation for the Real World (**Best Paper Award**)
C. Agia, R. Sinha, J. Yang, R. Antonova, M. Pavone, H. Nishimura, M. Itkina, and J. Bohg
<https://arxiv.org/abs/2506.19121>
- [4] **Real-Time Out-of-Distribution Failure Prevention via Multi-Modal Reasoning**
Conference on Robot Learning (CoRL), 2025 (**Oral Presentation, <16.0%**)
M. Ganai, R. Sinha[†], **C. Agia**[†], D. Morton, and M. Pavone
<https://arxiv.org/abs/2505.10547>
- [5] **RoboMonkey: Scaling Test-Time Sampling and Verification for Vision-Language-Action Models**
Conference on Robot Learning (CoRL), 2025
J. Kwok, **C. Agia**[†], R. Sinha[†], M. Foutter, S. Li, I. Stoica, A. Mirhoseini, and M. Pavone
<https://arxiv.org/abs/2506.17811>
- [6] **Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics**
International Conference on Robotics and Automation (ICRA), 2025
CoRL 2024 Workshop on Learning Effective Abstractions for Planning (**Oral Presentation**)
Y. Huang, **C. Agia**, J. Wu, T. Hermans, and J. Bohg
<https://arxiv.org/abs/2408.14769>

- [7] **Unpacking Failure Modes of Generative Policies: Runtime Monitoring of Consistency and Progress**
Conference on Robot Learning (CoRL), 2024
CoRL 2024 Workshop on Safe and Robust Robot Learning (Oral Presentation)
C. Agia, R. Sinha, J. Yang, Z. Cao, R. Antonova, M. Pavone, and J. Bohg
<https://arxiv.org/abs/2410.04640>
- [8] **Text2Interaction: Establishing Safe and Preferable Human-Robot Interaction**
Conference on Robot Learning (CoRL), 2024
J. Thumm, **C. Agia**, M. Pavone, and M. Althoff
<https://arxiv.org/abs/2408.06105>
- [9] **RoboMorph: In-Context Meta-Learning for Robot Dynamics Modeling**
International Conference on Informatics in Control, Automation and Robotics (ICINCO), 2024
M. Bazzi, A. Shahid, **C. Agia**, J. Alora, M. Forgione, D. Piga, F. Braghin, M. Pavone, and L. Roveda
<https://arxiv.org/abs/2409.11815>
- [10] **Real-Time Anomaly Detection and Reactive Planning with Large Language Models**
Robotics: Science and Systems (RSS), 2024 (Outstanding Paper Award, <0.1%)
R. Sinha, A. Elhafsi, **C. Agia**, M. Foutter, E. Schmerling, and M. Pavone
<https://arxiv.org/abs/2407.08735>
- [11] **DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset**
Robotics: Science and Systems (RSS), 2024
DROID Dataset Team
<https://arxiv.org/abs/2403.12945>
- [12] **Open X-Embodiment: Robotic Learning Datasets and RT-X Models**
International Conference on Robotics and Automation (ICRA), 2024 (Best Paper Award, <0.05%)
CoRL 2023 Workshop Towards Generalist Robots (Oral Presentation)
Open X-Embodiment Collaboration
<https://arxiv.org/abs/2310.08864>
- [13] **Modeling Considerations for Developing Deep Space Autonomous Spacecraft and Simulators**
IEEE Aerospace Conference (AeroConf), 2024
C. Agia, G. C. Vila, S. Bandyopadhyay, D. S. Bayard, K. Cheung, C. H. Lee, E. Wood, I. Aenishanslin, S. Ardito, L. Fesq, M. Pavone, and I. A. D. Nesnas
<https://arxiv.org/abs/2401.11371>
- [14] **Text2Motion: From Natural Language Instructions to Feasible Plans**
Autonomous Robots (AuRo), Special Issue: Large Language Models in Robotics, 2023
K. Lin[†], **C. Agia[†]**, T. Migimatsu, M. Pavone, and J. Bohg
<https://arxiv.org/abs/2303.12153>
- [15] **Semantic Anomaly Detection with Large Language Models**
Autonomous Robots (AuRo), Special Issue: Large Language Models in Robotics, 2023
A. Elhafsi, R. Sinha, **C. Agia**, E. Schmerling, I. A. D. Nesnas, and M. Pavone
<https://arxiv.org/abs/2305.11307>
- [16] **STAP: Sequencing Task-Agnostic Policies**
International Conference on Robotics and Automation (ICRA), 2023
C. Agia[†], T. Migimatsu[†], J. Wu, and J. Bohg
<https://arxiv.org/abs/2210.12250>
- [17] **Taskography: Evaluating robot task planning over large 3D scene graphs**
Conference on Robot Learning (CoRL), 2022
C. Agia[†], K. M. Jatavallabhula[†], M. Khodeir, O. Miksik, V. Vineet, M. Mukadam, L. Paull, and F. Shkurti
<https://arxiv.org/abs/2207.05006>
- [18] **Latent Attention Augmentation for Robust Autonomous Driving Policies**
International Conference on Intelligent Robots and Systems (IROS), 2021
R. Cheng[†], **C. Agia[†]**, F. Shkurti, D. Meger, and G. Dudek
<https://ieeexplore.ieee.org/document/9636449>

- [19] **S3CNet: A Sparse Semantic Scene Completion Network for LiDAR Point Clouds**
Conference on Robot Learning (CoRL), 2021
 R. Cheng[†], **C. Agia[†]**, Y. Ren, X. Li, and B. Liu
<https://arxiv.org/abs/2012.09242>
- [20] **Lightweight Semantic-Aided Localization With Spinning LiDAR Sensor**
IEEE Transactions on Intelligent Vehicles (T-IV), 2021
 Y. Ren, B. Liu, R. Cheng, and **C. Agia**
<https://ieeexplore.ieee.org/document/9495210>
- [21] **A Sim-to-Real Pipeline for Deep Reinforcement Learning for Autonomous Robot Navigation in Cluttered Rough Terrain**
IEEE Robotics and Automation Letters (RA-L), 2021
 H. Hu, K. Zhang, A. H. Tan, M. Ruan, **C. Agia**, and G. Nejat
<https://ieeexplore.ieee.org/document/9468918>
- [22] **Depth Prediction for Monocular Direct Visual Odometry**
Conference on Computer and Robot Vision (CRV), 2020
 R. Cheng, **C. Agia**, D. Meger, and G. Dudek
<https://ieeexplore.ieee.org/document/9108693>

Pre-Prints

- [1] **Behavior Graphs: Unsupervised Graph Structures for Interpreting Robot Data Flywheels**
In submission, 2026
 E. Bauer, J. Thumm, **C. Agia**, R. Sinha, C. Xu, H. Nishimura, J. Bohg, M. Pavone, and M. Itkina

Thesis / Dissertation Papers

- [1] **C. Agia. Deployment-Time Reliability of Learned Robot Policies.** Doctoral Dissertation. Stanford, CA: Stanford University, 2026. <https://arxiv.org/abs/2603.11400>
- [2] **C. Agia. Contextual Graph Representations for Task-driven 3D Perception and Planning.** Undergraduate Dissertation. Toronto, ON: University of Toronto, 2021. <https://arxiv.org/abs/2603.26685>

INVITED TALKS

RPL, UT Austin. Data-centric understanding of policy performance with influence functions (Talk)	2025-10
IEEE CASE. Data-centric understanding of policy performance with influence functions	2025-08
Toyota Research Institute. Data-centric understanding of policy performance with influence functions	2025-07
NASA University Leadership Initiative. Unpacking failure modes of generative policies	2024-10
SystemX Conference, Stanford University. Learning to solve long-horizon tasks (Talk)	2023-11
RVL, University of Toronto. Task and motion planning with skills and language models	2023-07
FAIR, Meta. Taskography: Evaluating robot task planning over large 3D scene graphs	2021-07
MSR, Microsoft. Robot task planning in structured world models	2021-07

ACADEMIC SERVICE AND TEACHING

Workshop Organization

1st Workshop on Making Sense of Data in Robotics @ CoRL 2025 ([Site](#)) (**Lead Organizer**)
 2nd Workshop on Out-of-Distribution Generalization in Robotics @ RSS 2025 ([Site](#))

Peer Reviewing

Journal Service: IJRR, RA-L
 Conference Service: RSS, CoRL, CVPR, ICRA, IROS, ISER, L4DC, AeroConf

Committee Service

Ph.D. Admissions Committee, CS & AA @ Stanford

Teaching

2× Stanford CS137A: Principles of Robot Autonomy 1	2023, 2024
1× Stanford CS237B: Principles of Robot Autonomy 2	2025
1× Stanford Test of AI & Emerging Technologies in service of the US Air Force (News)	2026

Research Mentorship

Ph.D. Students

Erik Bauer (Ph.D. CS @ Stanford)	2025 – 2026
Jacky Kwok (Ph.D. CS @ Stanford)	2024 – 2025
Milan Ganai (Ph.D. CS @ Stanford)	2024 – 2025
Hugo Buurmeijer (Ph.D. AA @ Stanford)	2024 – 2025

Undergraduate & Masters Students

Sreevardhan Sirigiri (B.S. Computing @ Sydney)	2025 – 2026
Kevin Lin (M.S. CS @ Stanford) → Ph.D. CS @ UT Austin	2022 – 2023
Bipasha Sen (M.S. by Research @ IIT Hyderabad) → Ph.D. EECS @ MIT	2021 – 2022

Visiting Researchers

Francesco Marchiori (Ph.D. CS @ Padua)	2024 – 2025
Yixuan Huang (Ph.D. Computing @ Utah) → Postdoc @ Princeton	2023 – 2024

PATENTS

Guiding Data Collection via Neurosymbolic Behavior Trees

U.S. Patent Pending, 2026

*E. Bauer, J. Thumm, **C. Agia**, R. Sinha, C. Xu, H. Nishimura, J. Bohg, M. Pavone, M. Itkina*

Curating Training Data for Robot Learning Based on Influence Functions

U.S. Patent Pending, 2025

***C. Agia**, R. Sinha, J. Yang, R. Antonova, H. Nishimura, M. Pavone, M. Itkina, J. Bohg*

Systems and Methods for Generating a Road Surface Semantic Segmentation Map from a Sequence of Point Clouds

U.S. Patent No. 12,008,762 (App. No. 17/676,131), USPTO, 2022 ([Google Patents](#))

***C. Agia**, R. Cheng, Y. Ren, B. Liu*

Methods and Systems for Semantic Scene Completion for Sparse 3D Data

U.S. Patent No. 12,079,970 (App. No. 17/492,261), USPTO, 2022 ([Google Patents](#))

*R. Cheng, **C. Agia**, Y. Ren, B. Liu*

COMMUNITY SERVICE AND LEADERSHIP

Canadian Student Association , Stanford University	2024-01 – 2025-01
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Co-founder and President of Stanford's Canadian Student Association

Stanford AI Salon , Stanford University	2021-09 – 2022-09
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Lead Organizer of Stanford's AI Salon, a platform facilitating open-ended discussion between graduate students, industry, and academic leaders on contemporary ML & AI topics and their societal implications

Stanford CS Mentorship Program , Stanford University	2021-09 – 2023-09
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Advising students from underrepresented and minority groups to lead fruitful careers in computer science research

Frosh Scholars Mentorship Program , Stanford University	2021-09 – 2022-09
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Mentoring first generation college students towards balanced progress in academics, career, and well-being

Pro Bono Research Mentoring	2021-01 – Present
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Guided many undergraduate research students through applications at top graduate engineering schools

NSight Student Mentorship Program , University of Toronto	2018-09 – 2019-05
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Provided academic, social, and personal support to first and second year Engineering Science students

SKILLS

Languages: (*Proficient*) Python, C/C#/C++, MATLAB, Rust, L^AT_EX, Bash - (*Working*) Java, Assembly

Tools: Git, Linux/Unix, Unity, Docker, Wasmtime (WebAssembly), Kubernetes

Libraries: PyTorch, TensorFlow, ROS, NumPy, ml-agents, PCL, OpenCV, SciPy, scikit-learn, Pandas, Jupyter